

PRACTICAL-1

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```
satwika8@LAPTOP-H4K52LE0:~$ cd CPrograms
satwika8@LAPTOP-H4K52LE0:~/CPrograms$ gedit practical1.c &
[1] 626
satwika8@LAPTOP-H4K52LE0:~/CPrograms$ gcc practical1.c -o practical1
satwika8@LAPTOP-H4K52LE0:~/CPrograms$ ./practical1
Enter a Linux command: ls
```

```
Parent Process
Parent PID: 699
Child PID: 700
```

```
Child Process
Child PID: 700
Parent PID: 699
fork hello hello.c practical1 practical1.c
```

Child process completed.

```
satwika8@LAPTOP-H4K52LE0:~/CPrograms$ uname
Linux
[1]+  Done                  gedit practical1.c
satwika8@LAPTOP-H4K52LE0:~/CPrograms$ ./practical1
Enter a Linux command: uname
```

```
Parent Process
Parent PID: 920
Child PID: 923
```

```
Child Process
Child PID: 923
Parent PID: 920
Linux
```

Child process completed.

```
satwika8@LAPTOP-H4K52LE0:~/CPrograms/fork$ uname -a
Linux LAPTOP-H4K52LE0 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

```

satwika8@LAPTOP-H4K52LE0:~/CPrograms/fork$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-13420H
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    1
Socket(s):              1
Stepping:               2
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx
pdpe1gb rdtsc
a cx16 pcid
ssbd ibrs i
smap clflush
64b fsrm md_
Virtualization features:
  Virtualization:      VT-x
  Hypervisor vendor:    Microsoft
  Virtualization type:  full
Caches (sum of all):
  L1d:                 288 KiB (6 instances)
  L1i:                 192 KiB (6 instances)
  L2:                  7.5 MiB (6 instances)
  L3:                  12 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-11
Vulnerabilities:
  Gather data sampling: Not affected

```

```

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Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:          Not affected
  Indirect target selection: Not affected
  Itlb multihit:       Not affected
  L1tf:                Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:     Not affected
  Old microcode:       Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:            Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:   Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:          Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:          Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:               Not affected
  Tsa:                 Not affected
  Tsx async abort:     Not affected
  Vmscape:             Not affected
satwika8@LAPTOP-H4K52LE0:~/CPrograms/fork$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 356.9M 1 disk
sdb  8:16   0 159.4M 1 disk
sdc  8:32   0    2G 0 disk [SWAP]
sdd  8:48   0    1T 0 disk /mnt/wslg/distro
/

```

```
satwika8@LAPTOP-H4K52LE0:~/CPrograms/fork$ ps
```

```
PID TTY TIME CMD
566 pts/0 00:00:00 bash
962 pts/0 00:00:00 ps
```

```
satwika8@LAPTOP-H4K52LE0:~/CPrograms/fork$ top
```

```
top - 05:58:18 up 49 min, 2 users, load average: 0.12, 0.05, 0.01
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7772.6 total, 7169.3 free, 530.4 used, 225.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7242.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21736	13204	9800	S	0.0	0.2	0:00.87	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
56	root	19	-1	66892	19660	18516	S	0.0	0.2	0:00.38	systemd-journal
104	root	20	0	25236	6644	5164	S	0.0	0.1	0:00.63	systemd-udev
118	systemd+	20	0	21344	13292	11168	S	0.0	0.2	0:00.11	systemd-resolve
119	systemd+	20	0	91036	7940	6968	S	0.0	0.1	0:00.15	systemd-timesyn
179	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
180	message+	20	0	9596	5404	4676	S	0.0	0.1	0:00.15	dbus-daemon
204	root	20	0	18160	8908	7848	S	0.0	0.1	0:00.12	systemd-logind
219	syslog	20	0	222516	5976	4592	S	0.0	0.1	0:00.08	rsyslogd
233	root	20	0	3124	1968	1836	S	0.0	0.0	0:00.00	agetty
266	root	20	0	112820	25160	15208	S	0.0	0.3	0:00.13	unattended-upgr
361	root	20	0	3184	1108	908	S	0.0	0.0	0:00.00	SessionLeader
362	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.03	Relay(363)
363	root	20	0	6076	5364	3688	S	0.0	0.1	0:00.02	bash
364	root	20	0	6672	4504	3736	S	0.0	0.1	0:00.00	login
408	root	20	0	20332	11524	9392	S	0.0	0.1	0:00.11	systemd
409	root	20	0	21168	3492	1784	S	0.0	0.0	0:00.00	(sd-pam)
435	root	20	0	6080	5320	3700	S	0.0	0.1	0:00.00	bash
540	root	20	0	6960	4516	4040	S	0.0	0.1	0:00.00	su
543	satwika8	20	0	20344	11600	9456	S	0.0	0.1	0:00.12	systemd
544	satwika8	20	0	21168	3572	1848	S	0.0	0.0	0:00.00	(sd-pam)
566	satwika8	20	0	6596	5916	3672	S	0.0	0.1	0:00.06	bash
629	satwika8	20	0	9436	5280	4700	S	0.0	0.1	0:00.04	dbus-daemon
633	satwika8	20	0	382836	8092	7324	S	0.0	0.1	0:00.00	at-spi-bus-laun
640	satwika8	20	0	9304	5076	4648	S	0.0	0.1	0:00.00	dbus-daemon
641	satwika8	20	0	474480	16348	13628	S	0.0	0.2	0:00.07	xdg-desktop-por
646	satwika8	20	0	531408	7396	6656	S	0.0	0.1	0:00.01	xdg-document-po
650	satwika8	20	0	304156	6176	5668	S	0.0	0.1	0:00.00	xdg-permission-